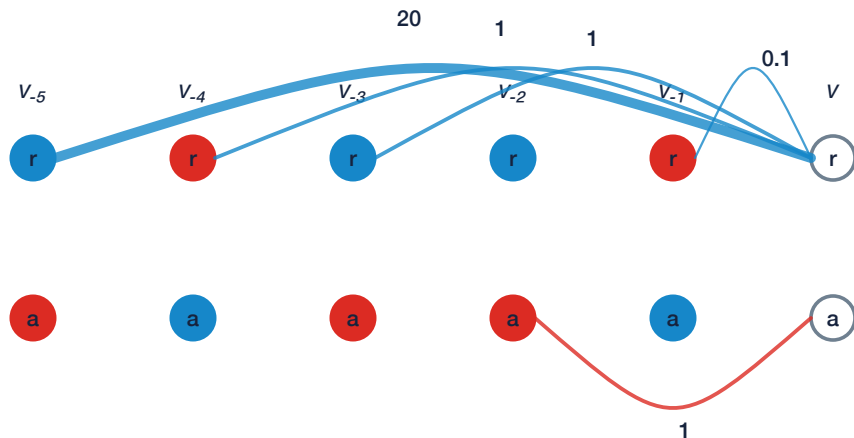


a Variant v receives weighted votes from up to k = 35 upstream variants



$$h_1(v) = 20 + 1 + 1 + 0.1 = 22.1 \quad h_2(v) = 1$$

v assigned to haplotype 1; tie $h_1 = h_2$ opens a new phase block (PS = position of first variant)

c Block formation

Variants are visited in coordinate order. Each assigned variant casts votes to its next k neighbours within 300 kb; a variant that receives no informative vote, or a tie, starts a new block. Blocks with a single variant are dropped. Haplotype labels are propagated along the block to produce phased GTs (0|1 / 1|0).

b Phasing entropy PE

$$PE(v) = - \sum p_i \log_2 p_i, \quad p_i = h_i / (h_1 + h_2)$$

votes for hap 1

22.1 : 1 PE = 0.257

3 : 1 PE = 0.811

votes for hap 2

4 : 3 PE = 0.985

5 : 5 PE = 1.000

$PE \geq 0.80$ triggers a GNN window (Supplementary Fig. 6)

Written to VCF as INFO/PE, INFO/H1, INFO/H2 for every phased variant.